Q2

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STAT 4C03

a)

```
df = read.csv("Pima.csv")
head(df)
```

```
##   pregnant glucose diastolic triceps insulin  bmi pedigree age diabetes
## 1        6     148        72      35       0 33.6    0.627  50        1
## 2        1      85        66      29       0 26.6    0.351  31        0
## 3        8     183        64       0       0 23.3    0.672  32        1
## 4        1      89        66      23      94 28.1    0.167  21        0
## 5        0     137        40      35     168 43.1    2.288  33        1
## 6        5     116        74       0       0 25.6    0.201  30        0
```

```
model.a = glm(diabetes~., data=df, family="binomial")
summary.a =summary(model.a)
summary.a
```

```
##
## Call:
## glm(formula = diabetes ~ ., family = "binomial", data = df)
##
## Coefficients:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept) -8.4046964  0.7166359 -11.728  < 2e-16 ***
## pregnant     0.1231823  0.0320776   3.840 0.000123 ***
## glucose      0.0351637  0.0037087   9.481  < 2e-16 ***
## diastolic   -0.0132955  0.0052336  -2.540 0.011072 *
## triceps      0.0006190  0.0068994   0.090 0.928515
## insulin     -0.0011917  0.0009012  -1.322 0.186065
## bmi          0.0897010  0.0150876   5.945 2.76e-09 ***
## pedigree     0.9451797  0.2991475   3.160 0.001580 **
## age          0.0148690  0.0093348   1.593 0.111192
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
```

```
##
##     Null deviance: 993.48  on 767  degrees of freedom
## Residual deviance: 723.45  on 759  degrees of freedom
## AIC: 741.45
##
## Number of Fisher Scoring iterations: 5
```

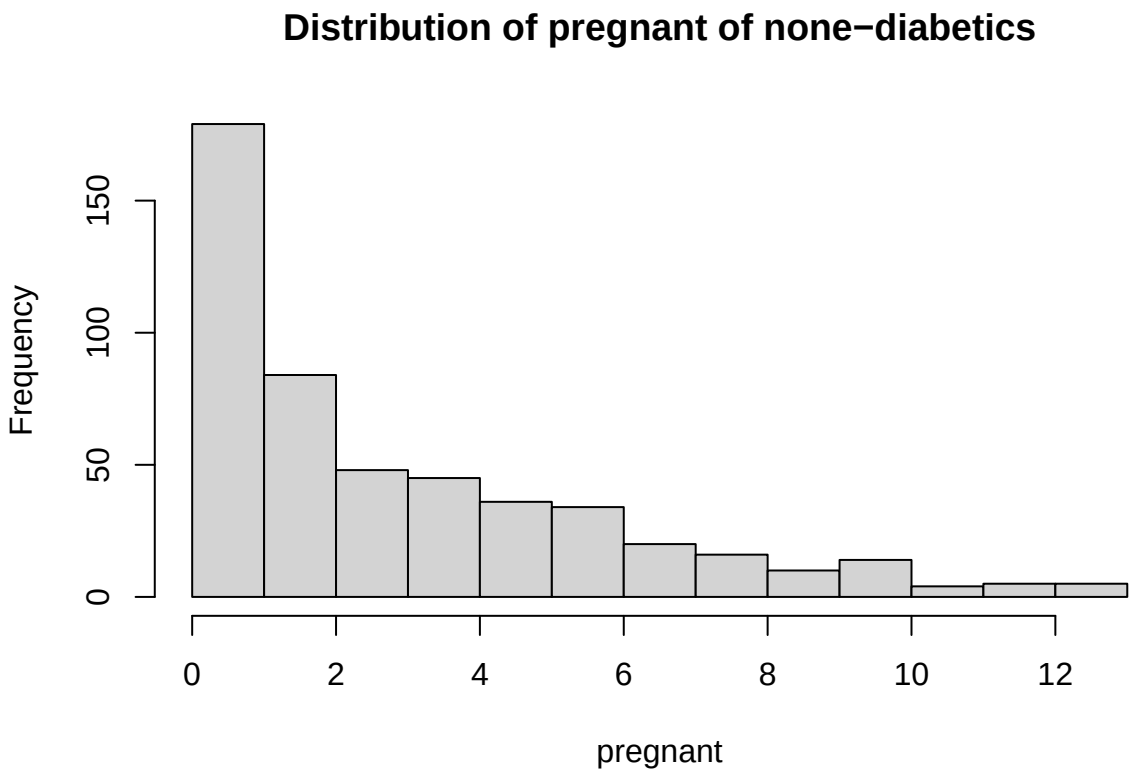
Report on the model analysis:

The results show that glucose, BMI, diabetes pedigree function and number of pregnancies are significant predictors of diabetes ($p < 0.05$), with glucose having the strongest effect. Diastolic blood pressure is also statistically significant after adjustment for the other variables. In contrast, triceps, insulin, and age are not significant in the full model. Overall, the model provides a substantial improvement over the null model.

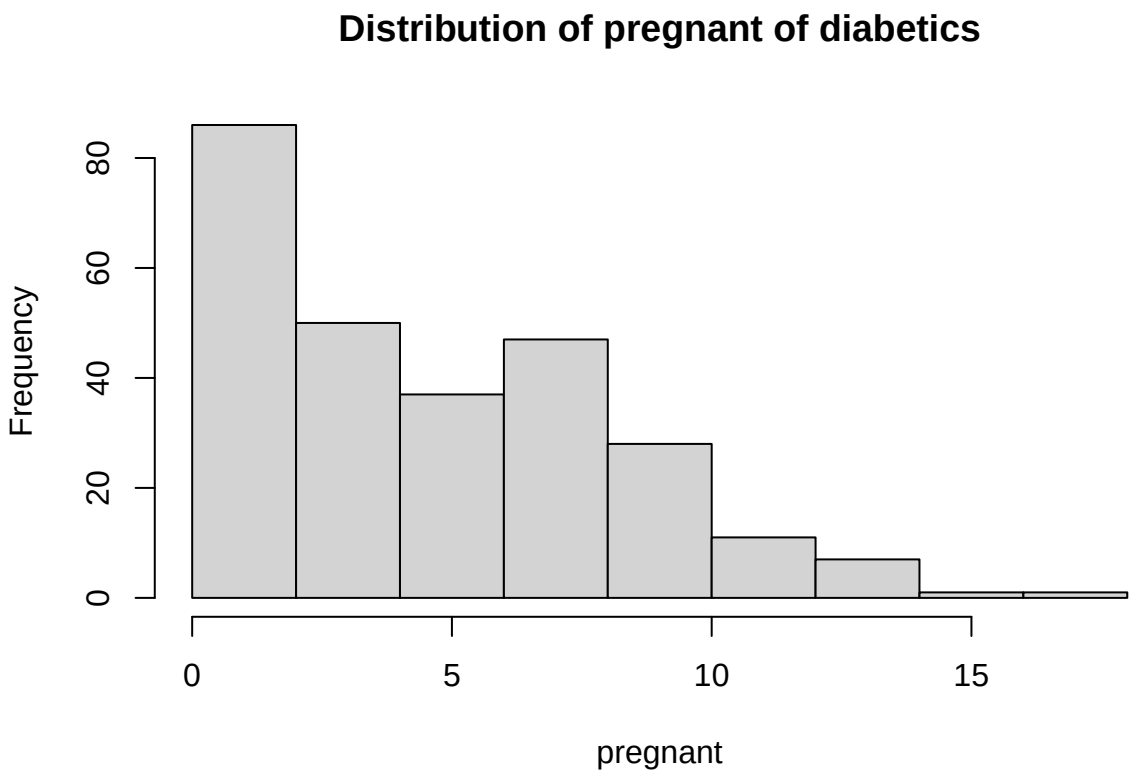
b)

```r
# We split the data based on the response variable
df_0 = df[df$diabetes==0,]
df_1 = df[df$diabetes==1,]
```

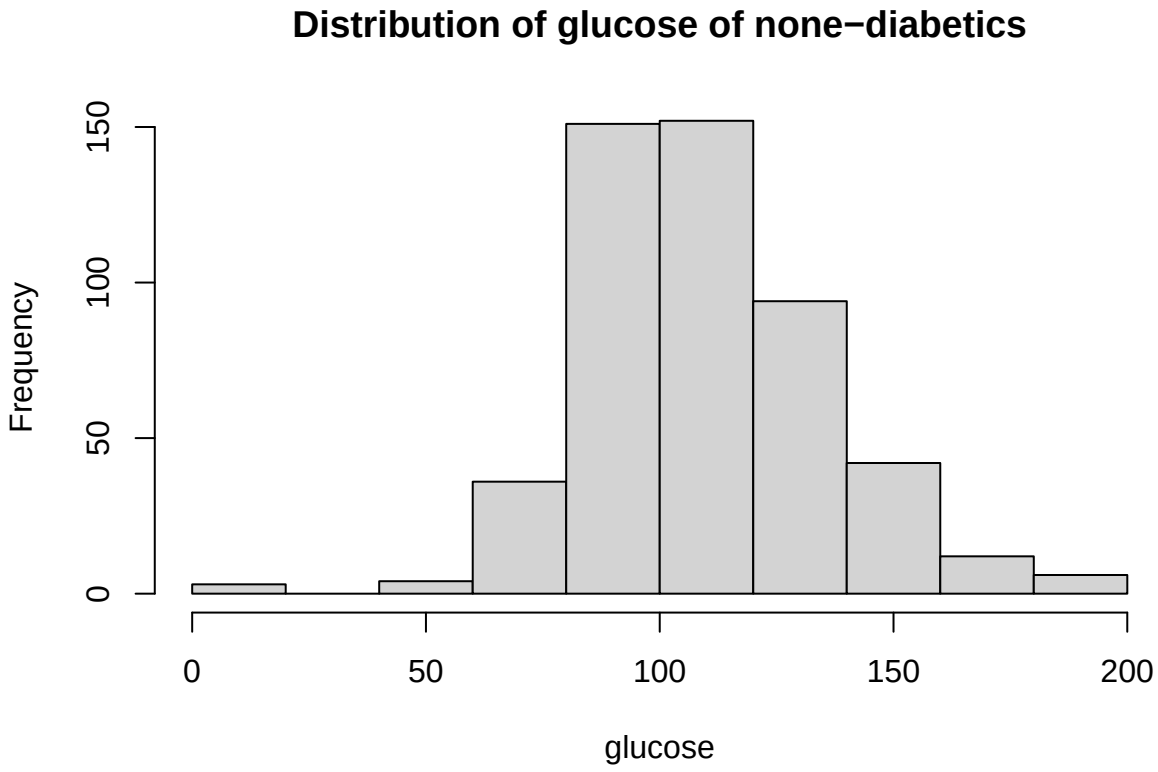
```r
for(i in 1:8){
  hist(df_0[,i], xlab=paste(colnames(df)[i]), main=sprintf("Distribution of %s of none-diabetics", colna
  cat("\\newpage")
  hist(df_1[,i], xlab=paste(colnames(df)[i]), main=sprintf("Distribution of %s of diabetics", colnames(
  cat("\\newpage")
}
```



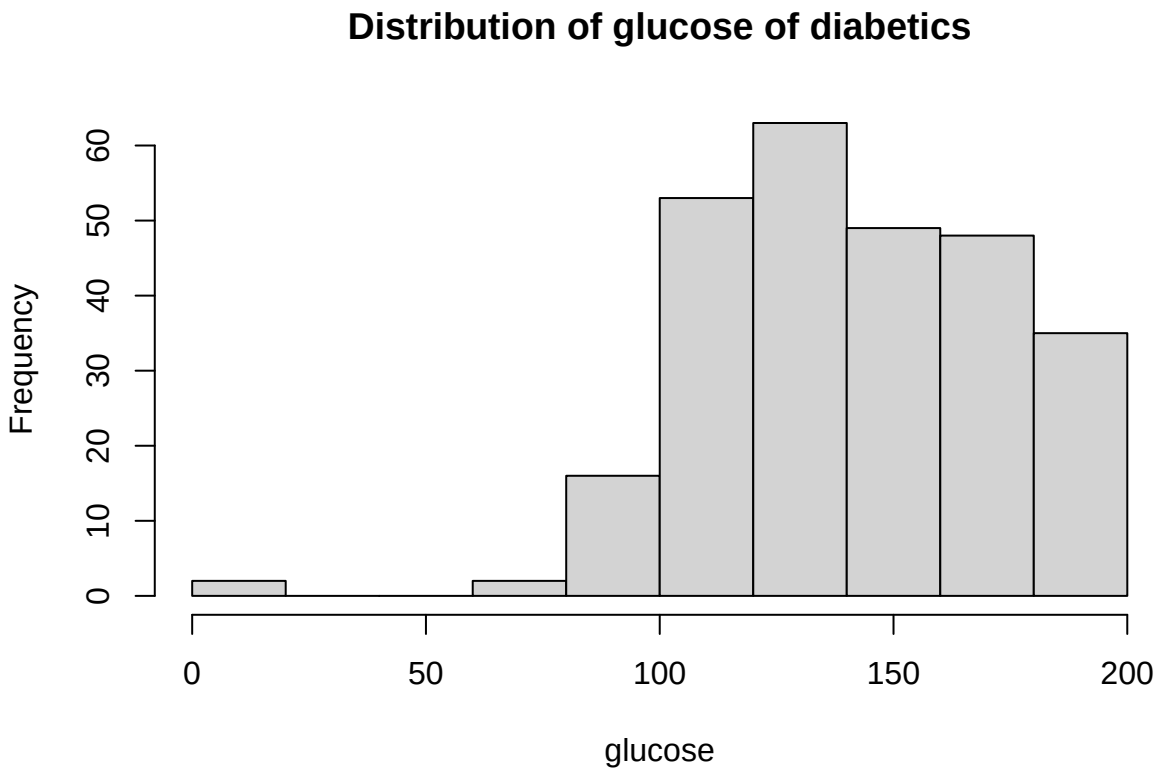
```
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```



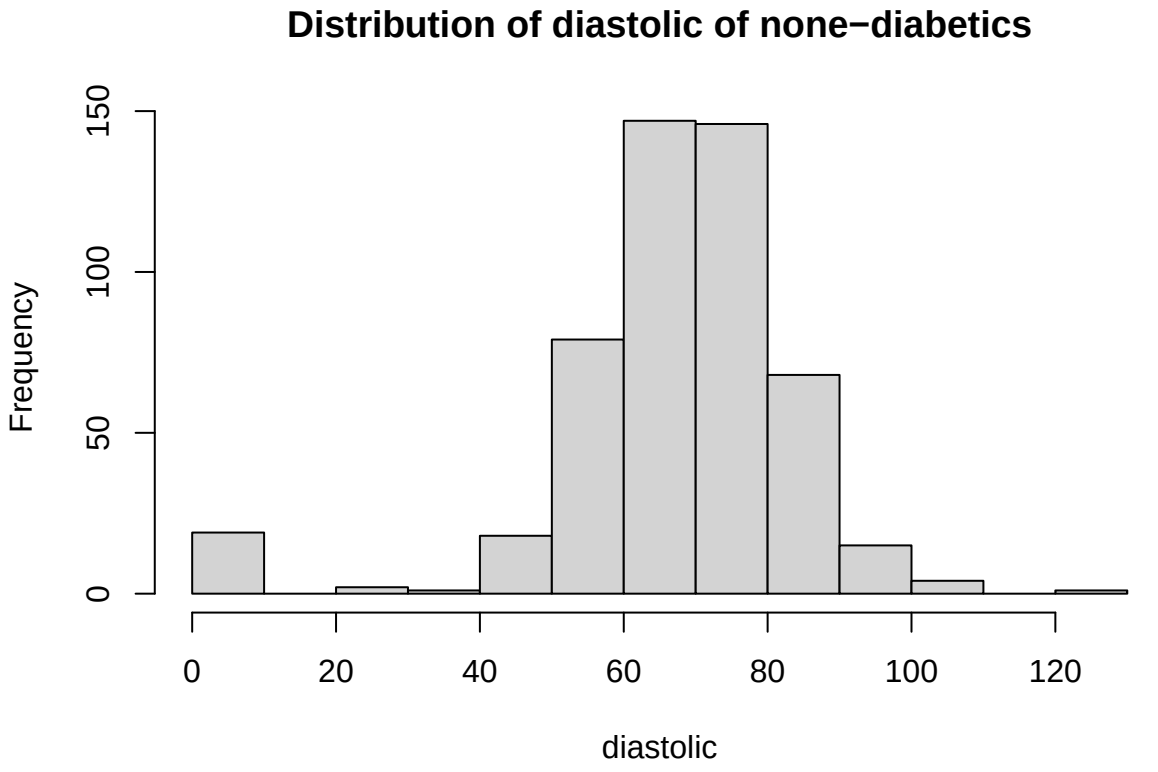

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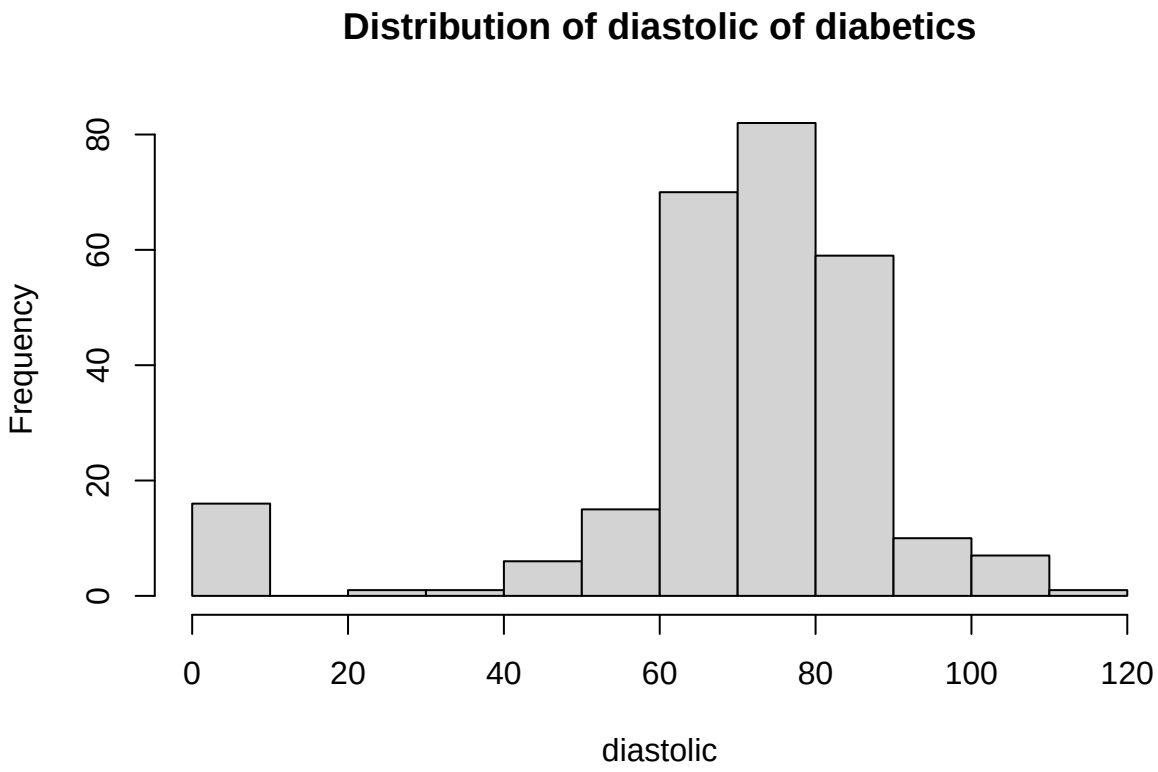
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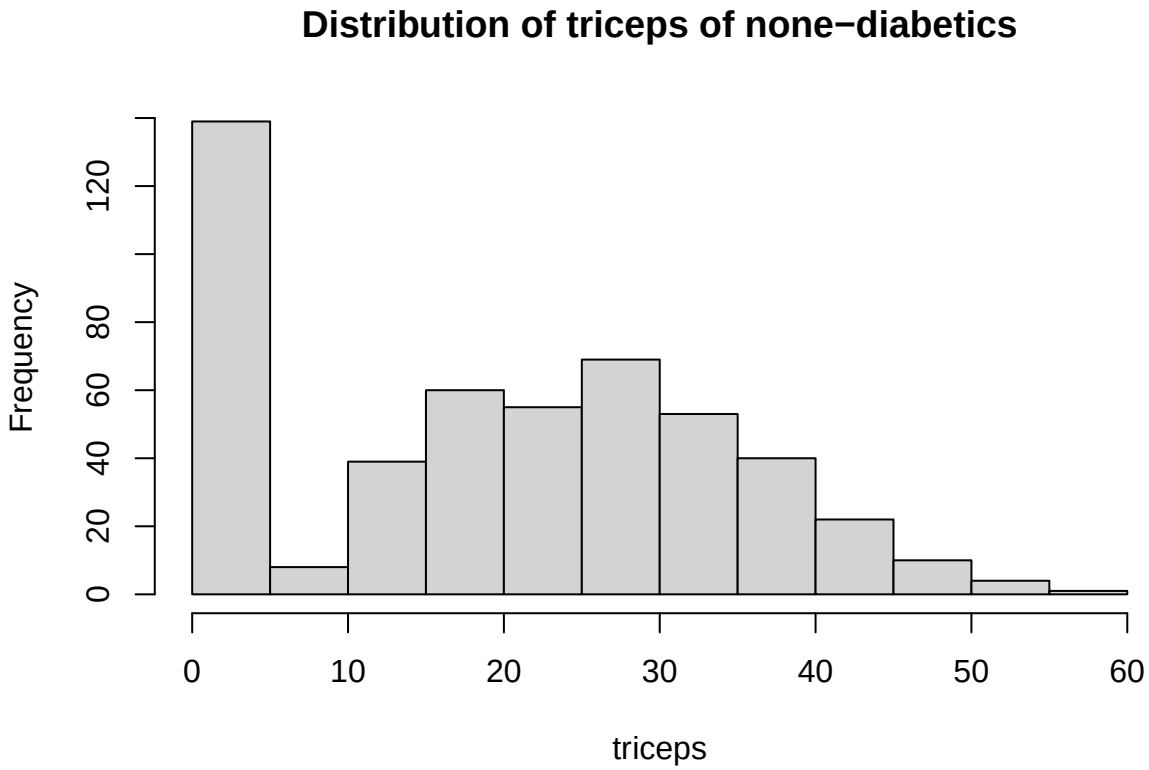

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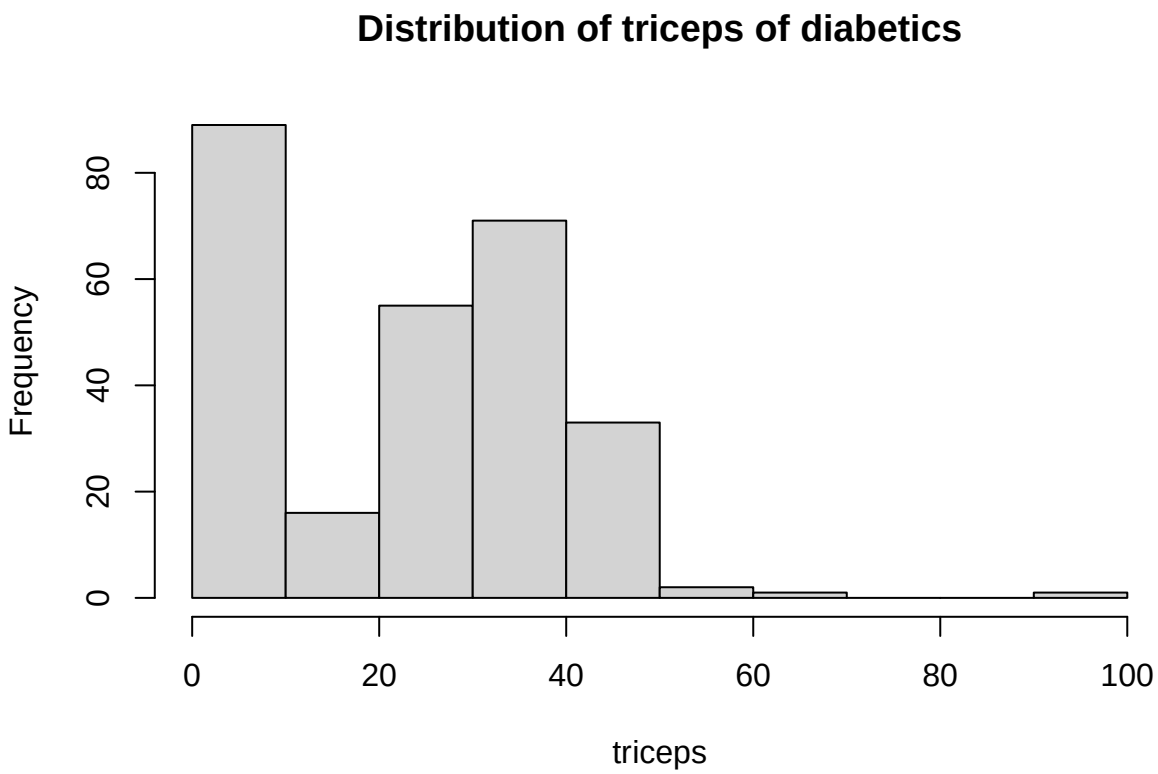
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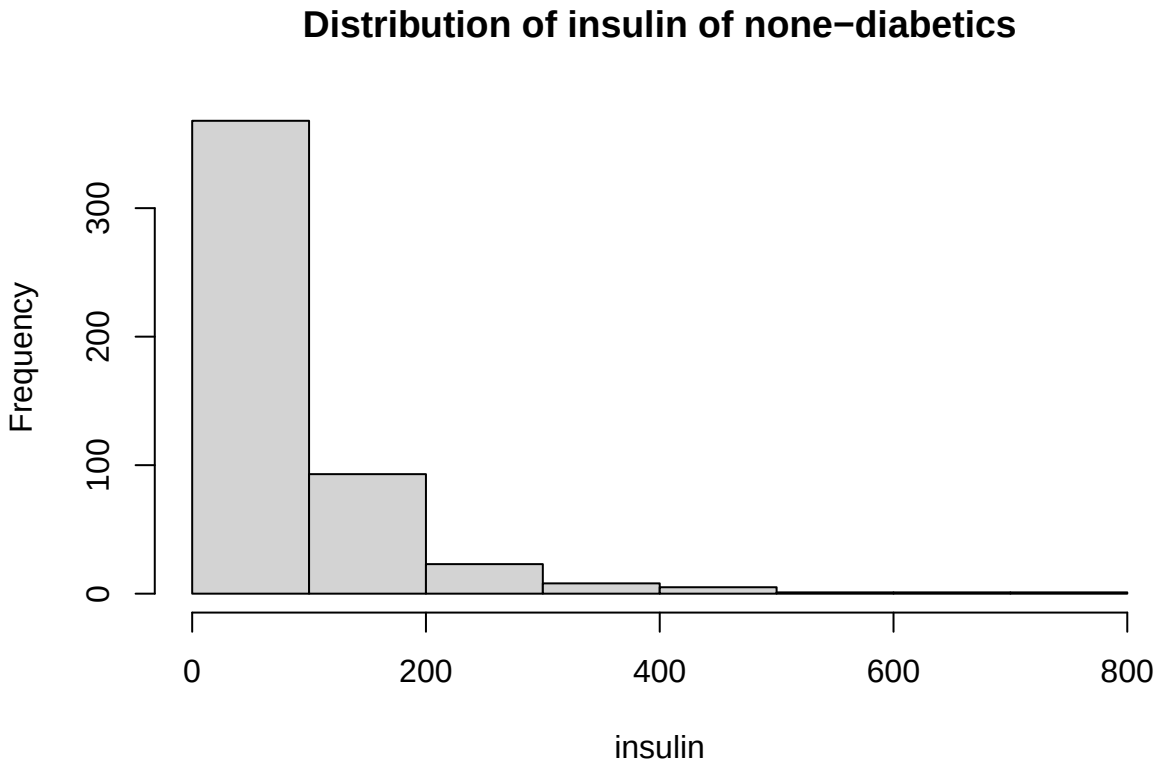

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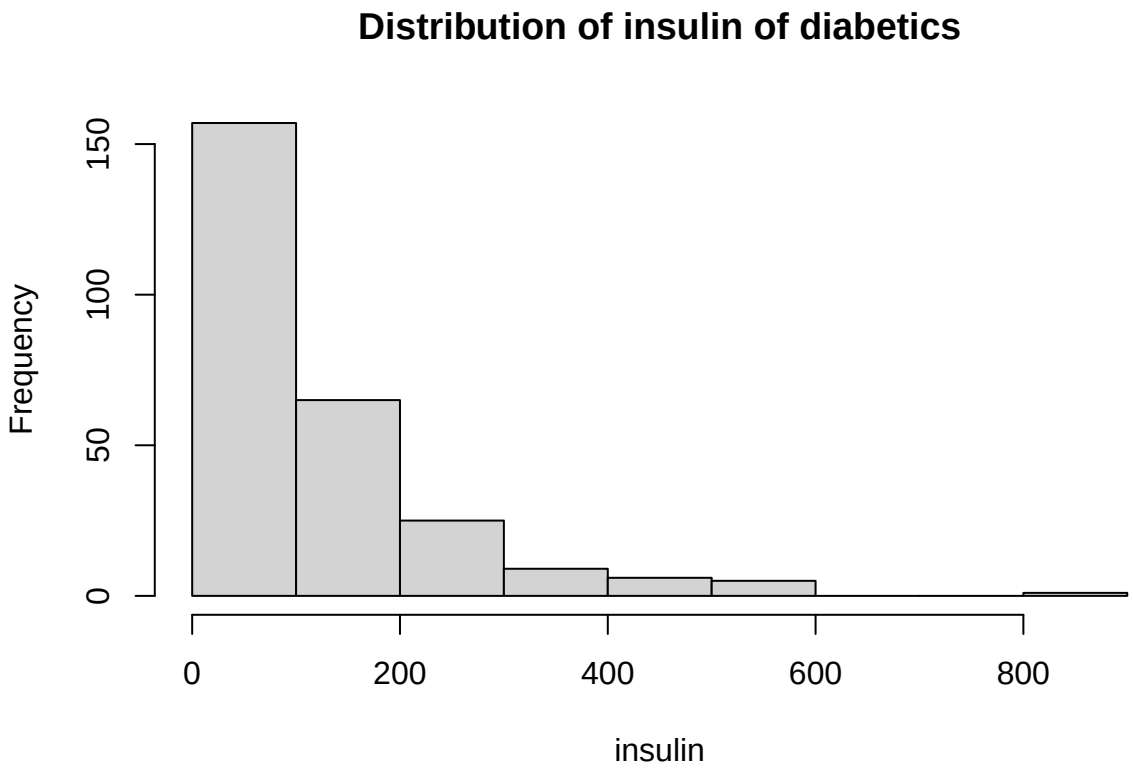
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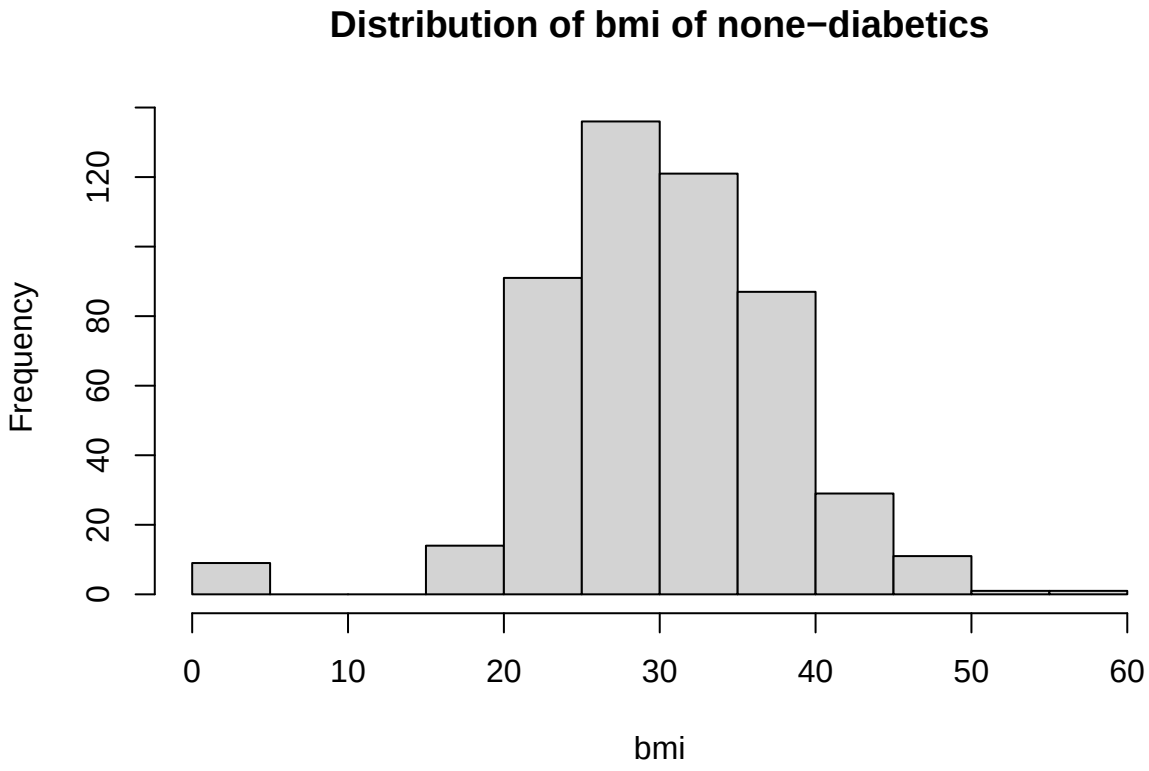

```
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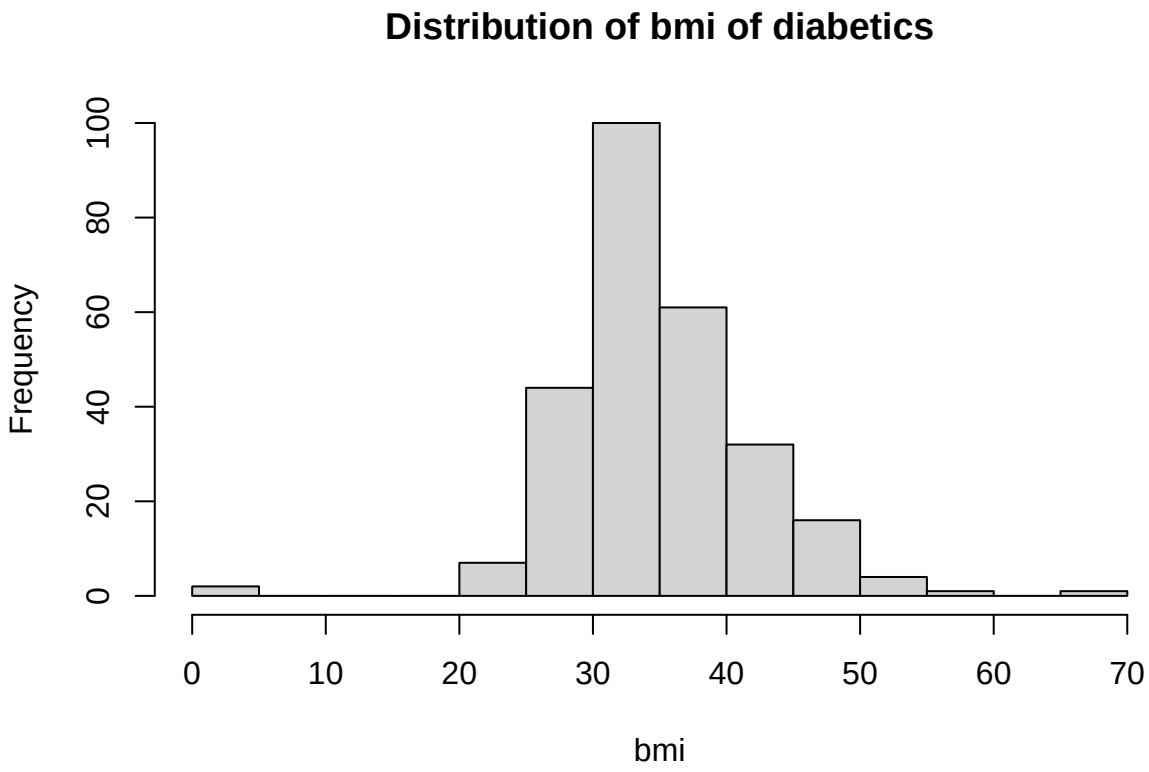
```
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```



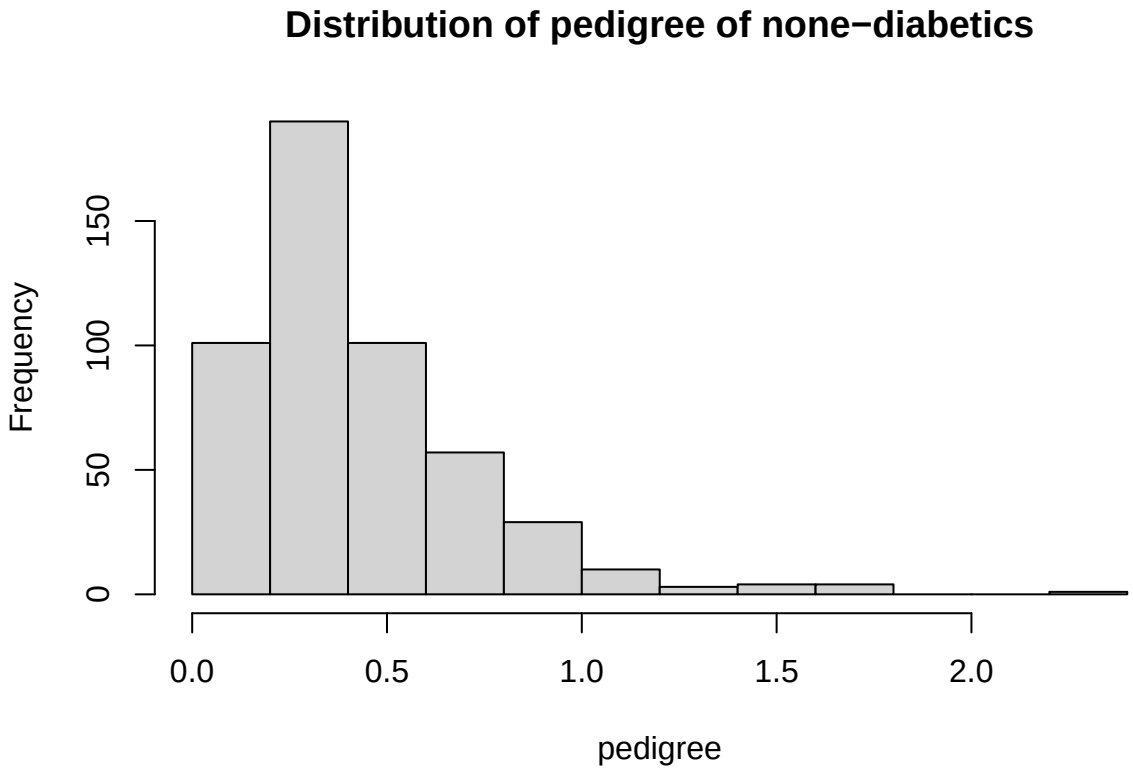

```
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```



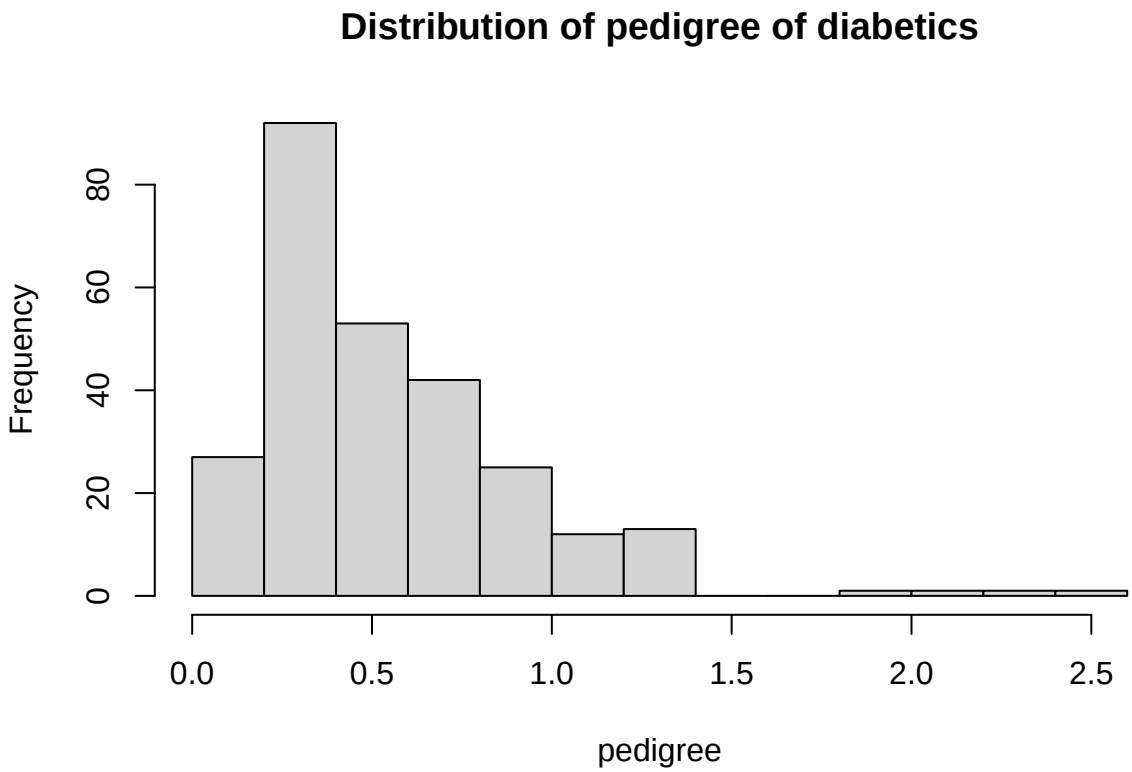
```
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```



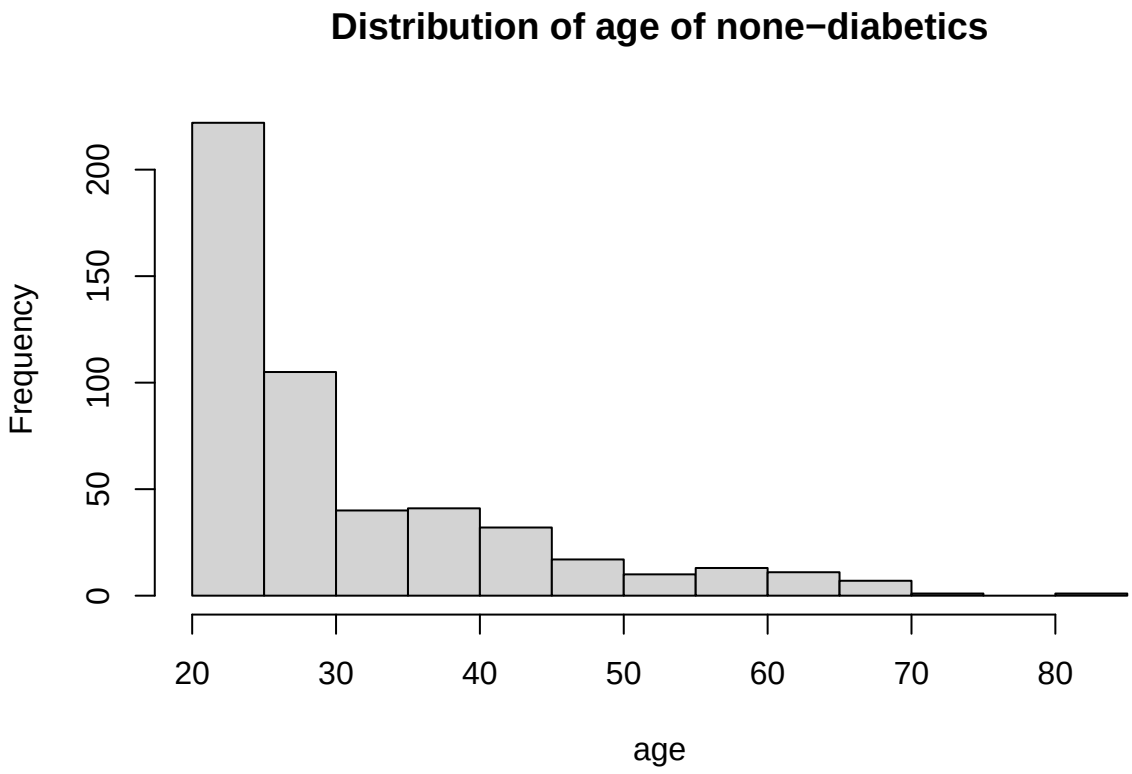

```
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```



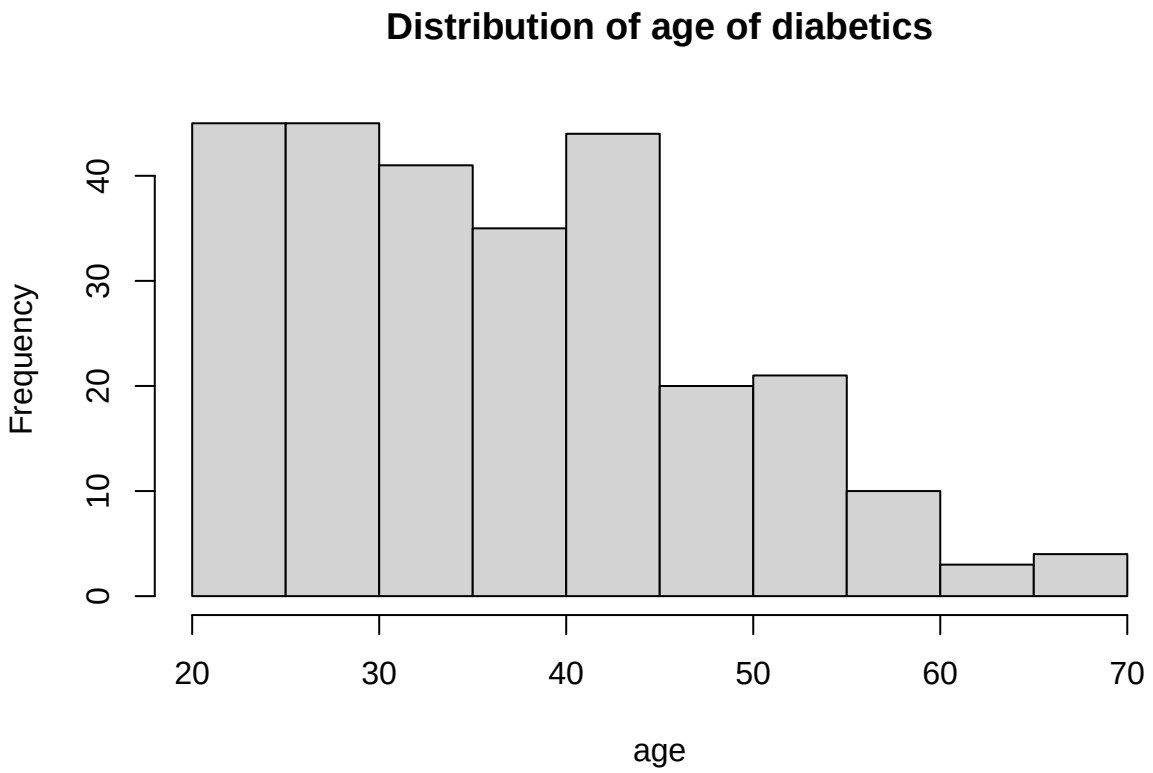
```
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```
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```



```
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```




```
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```

Analysis of the distributions:

Most of the distributions are relatively similar across the two groups, with only slight differences. However, for glucose, the distribution for diabetic individuals is clearly shifted to the right, showing a higher frequency of individuals with elevated glucose levels. Regarding age, the distribution for diabetic individuals appears more evenly spread, whereas among non-diabetic individuals, higher frequencies are observed at younger ages.

c)

```r
# Correcting irrelevant zero values on covariates
none.zero.covars = c("age", "glucose", "diastolic", "triceps", "bmi", "insulin")
for(covs in none.zero.covars){
  df[df[[covs]]==0, covs] <- NA
}
```

```r
nrow(df) - sum(complete.cases(df))
```

```
## [1] 376
```

Only 376 individuals have no missing values at all.

d)

```
# Refitting the model
model.d = glm(diabetes~., data=na.omit(df), family="binomial")
summary.d =summary(model.d)
summary.d
```

```
##
## Call:
## glm(formula = diabetes ~ ., family = "binomial", data = na.omit(df))
##
## Coefficients:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept) -1.004e+01  1.218e+00  -8.246  < 2e-16 ***
## pregnant     8.216e-02  5.543e-02   1.482  0.13825
## glucose      3.827e-02  5.768e-03   6.635 3.24e-11 ***
## diastolic   -1.420e-03  1.183e-02  -0.120  0.90446
## triceps      1.122e-02  1.708e-02   0.657  0.51128
## insulin     -8.253e-04  1.306e-03  -0.632  0.52757
## bmi          7.054e-02  2.734e-02   2.580  0.00989 **
## pedigree     1.141e+00  4.274e-01   2.669  0.00760 **
## age          3.395e-02  1.838e-02   1.847  0.06474 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 498.10  on 391  degrees of freedom
## Residual deviance: 344.02  on 383  degrees of freedom
## AIC: 362.02
##
## Number of Fisher Scoring iterations: 5
```

Report on the model.d's difference with model.a:

Compared to model.a, the model fitted on the complete-case dataset (model.d) shows some differences in statistical significance. In model.d, only glucose, BMI and pedigree remain significant at the 5% level, while pregnant and diastolic blood pressure are no longer significant. This change is mainly due to the reduced sample size after removing missing observations, which decreases statistical power and increases standard errors. However, glucose remains the strongest predictor in both models, indicating a robust association with diabetes.

e)

```
# Included individuals
included = complete.cases(df)

# Proportion of diabetes on included
prop.diabetes.included = mean(df$diabetes[included])
```

```r
# Proportion of diabetes on excluded
prop.diabetes.excluded = mean(df$diabetes[!included])

paste(
  "Proportion of diabetes among included individuals:",
  round(prop.diabetes.included, 3),
  "| Proportion of diabetes among excluded individuals:",
  round(prop.diabetes.excluded, 3)
)
```

```
## [1] "Proportion of diabetes among included individuals: 0.332 | Proportion of diabetes among excluded
```

The proportion of individuals with diabetes is slightly lower among those included in the analysis (33%) than among those excluded due to missing data (36%). This suggests that inclusion in the model is associated with the response. Therefor, the missing data are not completely at random..

f)

```r
# Refitting the model
model.f = glm(diabetes ~ pregnant + glucose + diastolic + bmi + pedigree + age, data = na.omit(df), fami
summary.f =summary(model.f)
summary.f
```

```
##
## Call:
## glm(formula = diabetes ~ pregnant + glucose + diastolic + bmi +
##     pedigree + age, family = binomial, data = na.omit(df))
##
## Coefficients:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept) -9.9604796  1.1818764  -8.428  < 2e-16 ***
## pregnant     0.0840497  0.0550728   1.526 0.126971
## glucose      0.0364863  0.0049973   7.301 2.85e-13 ***
## diastolic   -0.0008002  0.0118034  -0.068 0.945949
## bmi          0.0785728  0.0215674   3.643 0.000269 ***
## pedigree     1.1492368  0.4250340   2.704 0.006854 **
## age          0.0346079  0.0181919   1.902 0.057121 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 498.10  on 391  degrees of freedom
## Residual deviance: 344.88  on 385  degrees of freedom
## AIC: 358.88
##
## Number of Fisher Scoring iterations: 5
```

```r
# Likelihood Ratio Test
anova(model.d, model.f)
```

```
## Analysis of Deviance Table
##
## Model 1: diabetes ~ pregnant + glucose + diastolic + triceps + insulin +
##     bmi + pedigree + age
## Model 2: diabetes ~ pregnant + glucose + diastolic + bmi + pedigree +
##     age
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1       383     344.02
## 2       385     344.88 -2 -0.85931   0.6507
```

The test yields a p-value of 0.6507, indicating that the reduction in model complexity does not significantly worsen the model fit.

```r
# Refitting the reduced model with as much data as possible
model.f = glm(diabetes ~ pregnant + glucose + diastolic + bmi + pedigree + age, data = df, family = bin
summary.f =summary(model.f)
summary.f
```

```
##
## Call:
## glm(formula = diabetes ~ pregnant + glucose + diastolic + bmi +
##     pedigree + age, family = binomial, data = df)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -8.962146   0.820892 -10.918  < 2e-16 ***
## pregnant     0.117863   0.033418   3.527  0.00042 ***
## glucose      0.035194   0.003605   9.763  < 2e-16 ***
## diastolic   -0.008916   0.008618  -1.035  0.30084
## bmi          0.090926   0.015740   5.777 7.61e-09 ***
## pedigree     0.960515   0.306415   3.135  0.00172 **
## age          0.016944   0.009834   1.723  0.08489 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 931.94  on 723  degrees of freedom
## Residual deviance: 672.86  on 717  degrees of freedom
##   (44 observations deleted due to missingness)
## AIC: 686.86
##
## Number of Fisher Scoring iterations: 5
```

A total of 44 observations were ommited due to missingness so the final model used 762 - 44 = **724 data poins**

g)

```r
# Comparing the mean diastolic blood pressure for postive and negative individuals
t.test(diastolic ~ diabetes, data = df)
```

```
##
##  Welch Two Sample t-test
##
## data:  diastolic by diabetes
## t = -4.6643, df = 504.72, p-value = 3.972e-06
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
## 95 percent confidence interval:
##  -6.316023 -2.572156
## sample estimates:
## mean in group 0 mean in group 1
##        70.87734        75.32143
```

The two-sample t-test indicates a significant difference in mean diastolic blood pressure between diabetic and non-diabetic individuals. However, in the logistic regression model, diastolic blood pressure is not statistically significant because the regression assesses its conditional effect while controlling for other covariates. The disagreement arises due to confounding and the difference between marginal and conditional associations.